KUMARAGURU COLLEGE OF TECHNOLOGY

Department of Artificial Intelligence and Data Science

DATA SCIENCE AND VISUALIZATION (24ADI204)

WEEK 6 REPORT

Team Formation & Data Hunting

Team 16: Youth Smoking and Drug Consumption Analysis

DATA-VIZ HACK-A-BIT 2026–2027

Team Member Name	Roll Number
Lakshmi Prabha M M	25BAD049

Feature Analysis – Scaling and Transformation

1. Week 6 Contribution

Studied and compared data distributions before and after scaling and transformation techniques.

2. Work Carried Out

Examined how scaling and transformation affect numerical variables in the dataset. Compared the original distributions with the transformed distributions to understand changes in data range, spread and shape. Studied how these techniques help prepare data for further statistical analysis.

3. Topics Researched / Studied

- **Feature Scaling**
- **Min-Max Scaling**
- **Standardization (Z-score Scaling)**
- **Data Transformation**
- **Log Transformation**
- **Square Root Transformation**
- **Box-Cox Transformation**
- **Yeo-Johnson Transformation**

- **Before-and-After Distribution Comparison**

4. Outcome of My Contribution

Understood the importance of scaling and transformation in data analysis. Learned how these techniques change the range and distribution of numerical features and help improve data consistency.

5. Reference

Project notebook: DSV Team 16 – Youth Smoking and Drug Consumption Analysis.